



GATE ME 90 Days Challenge

Day 11/90 · Practice Set with Detailed Solutions

SUBJECT

Strength of Materials

TODAY'S TOPIC

Torsion, Thin Cylinders & Springs

COUNTDOWN

99 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on torsion of solid and hollow shafts, power transmission, compound shafts, thin cylindrical and spherical shells, and close-coiled helical springs. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 (MCQ)

The polar section modulus Z_p of a hollow circular shaft of outer diameter D_o and inner diameter D_i is

(A) $\pi(D_o^4 - D_i^4)/(32 D_o)$

(B) $\pi(D_o^4 - D_i^4)/(16 D_o)$

(C) $\pi(D_o^3 - D_i^3)/16$

(D) $\pi(D_o^4 - D_i^4)/(16 D_i)$

Q2 (NAT)

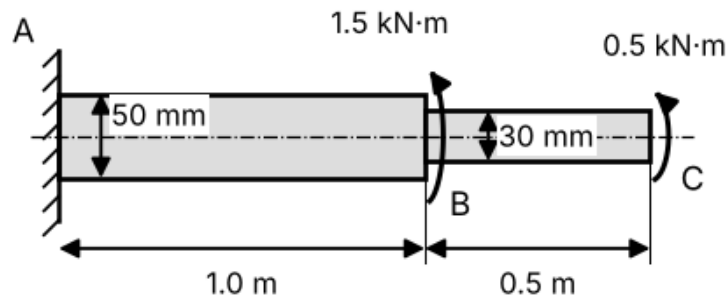
A solid shaft of 60 mm diameter runs at 180 rpm. If the maximum shear stress is limited to 50 MPa, the power it can transmit (in kW) is _____

Q3 (NAT)

A solid steel shaft must transmit 75 kW at 300 rpm. The shear stress must not exceed 60 MPa and the angle of twist must not exceed 0.25° per metre length ($G = 80 \text{ GPa}$). The minimum shaft diameter (in mm) is _____

Q4 (MCQ)

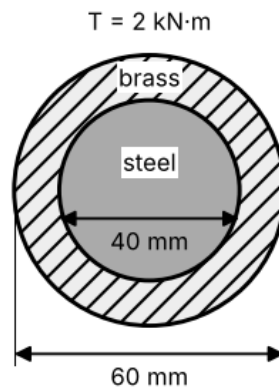
The stepped steel shaft ABC shown is fixed at A. AB is 50 mm in diameter and 1.0 m long; BC is 30 mm in diameter and 0.5 m long. Torques of 1.5 kN·m at B and 0.5 kN·m at C act in the same sense. The maximum shear stress in the shaft is



- (A) 81.5 MPa
(B) 61.1 MPa
(C) 94.3 MPa
(D) 377.3 MPa

Q5 (NAT)

A composite shaft is made of a solid steel core of 40 mm diameter rigidly bonded inside a brass tube of 60 mm outer diameter, as shown. The shaft carries a torque of 2 kN·m. With $G_{\text{steel}} = 80 \text{ GPa}$ and $G_{\text{brass}} = 40 \text{ GPa}$, the maximum shear stress in the brass (in MPa) is _____

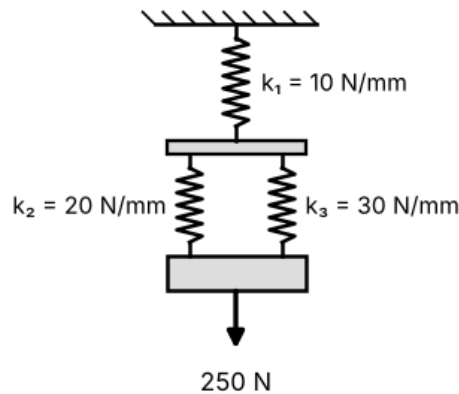
**Q6** (MCQ)

A shaft section carries a bending moment of 2 kN·m and a torque of 1.5 kN·m. The equivalent bending moment (maximum principal stress theory) is

- (A) 2.25 kN·m
(B) 2.50 kN·m
(C) 1.75 kN·m
(D) 3.50 kN·m

Q12 MCQ

The spring system shown carries a load of 250 N. Spring 1 ($k_1 = 10 \text{ N/mm}$) is connected in series with springs 2 ($k_2 = 20 \text{ N/mm}$) and 3 ($k_3 = 30 \text{ N/mm}$), which act in parallel. The deflection of the load is



- (A) 45.8 mm (B) 4.2 mm
(C) 25.0 mm (D) 30.0 mm

Q13 **NAT**

A close-coiled helical spring of 10 mm wire and 60 mm mean coil diameter ($G = 80 \text{ GPa}$) must deflect 25 mm under an axial load of 800 N. The number of active turns required (whole number) is _____

Q14 MCQ

The Wahl correction factor for a close-coiled helical spring with a spring index of 6 is closest to

- (A) 1.253 (B) 1.083
(C) 1.150 (D) 1.333

Q15 MCQ

A close-coiled helical spring of stiffness k is cut into three equal lengths, and the three pieces are placed side by side in parallel to carry a load. The stiffness of the arrangement is

- (A) $3k$ (B) $9k$
(C) $k/3$ (D) $k/9$

Answer Key

Q	Type	Answer
Q1	MCQ	(B) $\pi(D_o^4 - D_i^4)/(16 D_o)$
Q2	NAT	39.97 (accept 39.7 to 40.2)
Q3	NAT	91.4 (accept 91.0 to 91.8)
Q4	MCQ	(C) 94.3 MPa
Q5	NAT	39.38 (accept 39.0 to 39.8)
Q6	MCQ	(A) 2.25 kN·m
Q7	NAT	168.4 (accept 167.5 to 169.3)
Q8	MCQ	(D) 0.266 mm
Q9	NAT	570 (accept 565 to 575)
Q10	MCQ	(C) 1.92 MPa
Q11	NAT	1.18 (accept 1.17 to 1.19)
Q12	MCQ	(D) 30.0 mm
Q13	NAT	15 (calculated 14.47; round up)
Q14	MCQ	(A) 1.253
Q15	MCQ	(B) 9k

Detailed Solutions

Q1 Answer: (B) $\pi(D_o^4 - D_i^4)/(16 D_o)$

Concept: $Z_p = J / r_{\max}$, where $r_{\max}$ is the outer radius, because the shear stress is greatest at the outer surface.

$$J = \pi(D_o^4 - D_i^4)/32 \text{ and } r_{\max} = D_o/2$$

$$Z_p = J/(D_o/2) = \pi(D_o^4 - D_i^4)/(16 D_o), \text{ so } T = \tau_{\max} Z_p.$$

Exam tip: Option (C) is the classic trap: the section modulus of a hollow shaft is NOT the difference of the solid moduli ($\pi D_o^3/16 - \pi D_i^3/16$), because both parts must be referred to the same outer radius.

Q2 Answer: 39.97 (accept 39.7 to 40.2)

Concept: Torque capacity from the torsion formula, then $P = 2\pi NT/60$.

$$T = \pi d^3 \tau / 16 = \pi \times 60^3 \times 50 / 16 = 2.1206 \times 10^6 \text{ N}\cdot\text{mm} = 2120.6 \text{ N}\cdot\text{m}$$

$$P = 2\pi \times 180 \times 2120.6 / 60 = 39\,972 \text{ W} = \mathbf{39.97 \text{ kW}}$$

Exam tip: Keep units straight: T in N·m and N in rpm give P in watts. A slip with N·mm makes the answer 1000 times too large.

Q3 Answer: 91.4 (accept 91.0 to 91.8)

Concept: Design for strength and for stiffness separately; the larger diameter satisfies both.

$$T = 60P/(2\pi N) = 60 \times 75\,000/(2\pi \times 300) = 2387.3 \text{ N}\cdot\text{m}$$

$$\text{Strength: } d^3 = 16T/(\pi\tau) = 16 \times 2.3873 \times 10^6/(\pi \times 60) \rightarrow d = 58.7 \text{ mm}$$

$$\text{Stiffness: } \theta/L = 0.25^\circ \text{ per m} = 4.363 \times 10^{-6} \text{ rad/mm} = T/(GJ) \rightarrow J = 2.3873 \times 10^6/(80\,000 \times 4.363 \times 10^{-6}) = 6.839 \times 10^6 \text{ mm}^4$$

$$d^4 = 32J/\pi \rightarrow d = \mathbf{91.4 \text{ mm}}. \text{ Stiffness governs.}$$

Exam tip: Long transmission shafts are usually stiffness-limited. Always convert the twist limit to radians per mm before using $T/J = G\theta/L$.

Q4 Answer: (C) 94.3 MPa

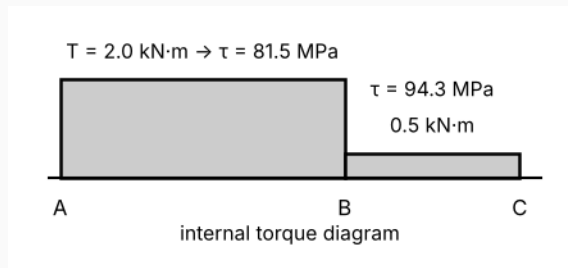
Concept: Shafts in series: find the internal torque in each segment from the part to the right of a cut, then apply $\tau = 16T/(\pi d^3)$ segment by segment.

$$\text{Segment BC: } T_{BC} = 0.5 \text{ kN}\cdot\text{m} \rightarrow \tau = 16 \times 0.5 \times 10^6/(\pi \times 30^3) = \mathbf{94.3 \text{ MPa}}$$

$$\text{Segment AB: } T_{AB} = 1.5 + 0.5 = 2.0 \text{ kN}\cdot\text{m} \rightarrow \tau = 16 \times 2 \times 10^6/(\pi \times 50^3) = 81.5 \text{ MPa}$$

So the thin segment BC governs: $\tau_{\max} = \mathbf{94.3 \text{ MPa}}$. (With $G = 80 \text{ GPa}$, the twist at C is $\theta_{AB} + \theta_{BC} = 0.0407 + 0.0393 = 0.0800 \text{ rad} = 4.59^\circ$.)

Exam tip: The largest torque does not always give the largest stress. Compare T/d^3 for each segment.



Q5 Answer: 39.38 (accept 39.0 to 39.8)

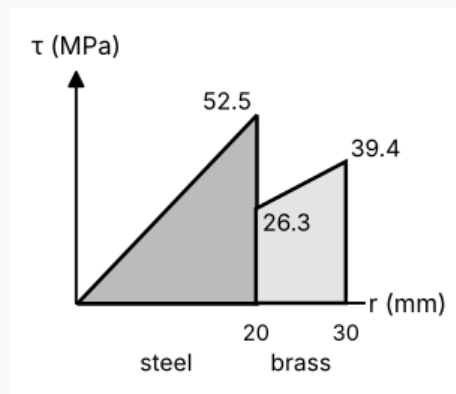
Concept: Shafts in parallel (statically indeterminate): both parts twist through the same angle, so torque divides in proportion to the torsional rigidity GJ .

$$J_s = \pi \times 40^4 / 32 = 2.513 \times 10^5 \text{ mm}^4; J_b = \pi(60^4 - 40^4) / 32 = 1.021 \times 10^6 \text{ mm}^4$$

$$G_s J_s = 2.011 \times 10^{10}, G_b J_b = 4.084 \times 10^{10} \text{ N}\cdot\text{mm}^2 \rightarrow T_b = 2 \times 4.084 / 6.095 = 1.340 \text{ kN}\cdot\text{m}, T_s = 0.660 \text{ kN}\cdot\text{m}$$

$$\tau_{b,\max} = T_b \times 30 / J_b = 1.340 \times 10^6 \times 30 / 1.021 \times 10^6 = \mathbf{39.38 \text{ MPa}}$$
 (steel: $0.660 \times 10^6 \times 20 / 2.513 \times 10^5 = 52.5 \text{ MPa}$)

Exam tip: Shear strain is continuous across the bond, but shear stress jumps at the interface by the ratio G_s/G_b (here $52.5 \rightarrow 26.3 \text{ MPa}$).

**Q6 Answer: (A) 2.25 kN·m**

Concept: Combined bending and torsion: $M_e = [M + \sqrt{(M^2 + T^2)}] / 2$ and $T_e = \sqrt{(M^2 + T^2)}$.

$$T_e = \sqrt{(2^2 + 1.5^2)} = 2.5 \text{ kN}\cdot\text{m}$$

$$M_e = (2 + 2.5) / 2 = \mathbf{2.25 \text{ kN}\cdot\text{m}}$$

Exam tip: $2.50 \text{ kN}\cdot\text{m}$ is T_e , which goes with the maximum shear stress theory. This topic returns in detail with shaft design.

Q7 Answer: 168.4 (accept 167.5 to 169.3)

Concept: In pure torsion the surface element is in pure shear, so the principal strains at $\pm 45^\circ$ are $\pm \gamma / 2$.

$$\gamma = 2\varepsilon_{45} = 2 \times 200 \times 10^{-6} = 4 \times 10^{-4}; \tau = G\gamma = 80\,000 \times 4 \times 10^{-4} = 32 \text{ MPa}$$

$$T = \pi d^3 \tau / 16 = \pi \times 80^3 \times 32 / 16 = 3.217 \times 10^6 \text{ N}\cdot\text{mm} = 3217 \text{ N}\cdot\text{m}$$

$$P = 2\pi \times 500 \times 3217 / 60 = \mathbf{168.4 \text{ kW}}$$

Exam tip: Using $\gamma = \varepsilon_{45}$ (forgetting the factor 2) halves the answer to 84.2 kW , a favourite distractor.

Q8 Answer: (D) 0.266 mm

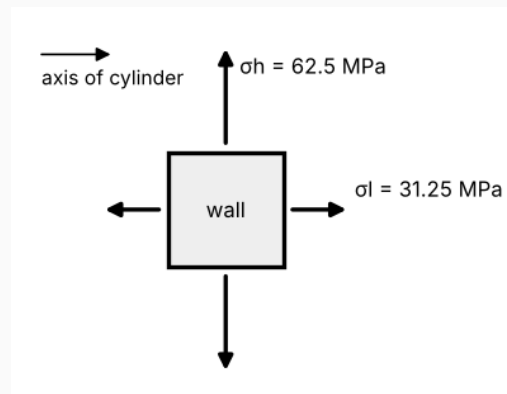
Concept: The hoop strain controls the diameter change: $\epsilon_h = (\sigma_h - \nu\sigma_l)/E$, with $\sigma_h = pd/2t$ and $\sigma_l = pd/4t$.

$$\sigma_h = 1.5 \times 1000/(2 \times 12) = 62.5 \text{ MPa}; \sigma_l = 31.25 \text{ MPa}$$

$$\epsilon_h = (62.5 - 0.3 \times 31.25)/200\,000 = 2.656 \times 10^{-4}$$

$$\Delta d = \epsilon_h d = 2.656 \times 10^{-4} \times 1000 = \mathbf{0.266 \text{ mm}}$$

Exam tip: 0.313 mm ignores the Poisson effect of the longitudinal stress. In compact form, $\Delta d = (pd^2/4tE)(2 - \nu)$.

**Q9 Answer: 570 (accept 565 to 575)**

Concept: In a thin sphere the wall stress $pd/4t$ acts equally in all directions, so $\epsilon_v = 3\epsilon = 3(\sigma/E)(1 - \nu)$.

$$\sigma = 1.6 \times 1200/(4 \times 8) = 60 \text{ MPa}$$

$$\epsilon_v = 3 \times (60/200\,000) \times 0.7 = 6.3 \times 10^{-4}$$

$$V = \pi d^3/6 = \pi \times 1.2^3/6 = 0.9048 \text{ m}^3 \rightarrow \Delta V = 6.3 \times 10^{-4} \times 0.9048 = 5.70 \times 10^{-4} \text{ m}^3 = \mathbf{570 \text{ cm}^3}$$

Exam tip: $1 \text{ m}^3 = 10^6 \text{ cm}^3 = 1000 \text{ litres}$. Unit conversion is where most wrong answers in this question come from.

Q10 Answer: (C) 1.92 MPa

Concept: The hoop stress is resisted by the longitudinal joint, the longitudinal stress by the circumferential joint. Check both and take the smaller pressure.

$$\text{Hoop: } pd/(2t) \leq \sigma \eta_l \rightarrow p = 2 \times 20 \times 120 \times 0.8/2000 = \mathbf{1.92 \text{ MPa}}$$

$$\text{Longitudinal: } pd/(4t) \leq \sigma \eta_c \rightarrow p = 4 \times 20 \times 120 \times 0.5/2000 = 2.40 \text{ MPa}$$

Safe pressure = smaller value = **1.92 MPa**

Exam tip: Matching the joint with the stress it carries is the whole question: longitudinal joint $\leftrightarrow$ hoop stress, circumferential joint $\leftrightarrow$ longitudinal stress.

Q11 Answer: 1.18 (accept 1.17 to 1.19)

Concept: $\epsilon_h = (\sigma_h - \nu\sigma_l)/E = (pd/4tE)(2 - \nu)$.

$$p = 4tE\epsilon_h/[d(2 - \nu)] = 4 \times 6 \times 200\,000 \times 250 \times 10^{-6}/(600 \times 1.7)$$

$$p = 1200/1020 = \mathbf{1.18 \text{ MPa}}$$

Exam tip: Ignoring Poisson's ratio would give 1.00 MPa; the longitudinal stress reduces the hoop strain, so the true pressure is higher.

Q12 Answer: (D) 30.0 mm

Concept: Parallel springs share the deflection (stiffnesses add); series springs share the load (flexibilities add).

Parallel pair: $k_{23} = 20 + 30 = 50 \text{ N/mm}$

Series with k_1 : $1/k = 1/10 + 1/50 \rightarrow k = 8.33 \text{ N/mm}$

$\delta = 250/8.33 = \mathbf{30.0 \text{ mm}}$ (= 25 mm in spring 1 + 5 mm in the parallel pair)

Exam tip: 45.8 mm treats all three in series and 4.2 mm all three in parallel. Read the connection from the figure before writing any formula.

Q13 Answer: 15 (calculated 14.47; round up)

Concept: $\delta = 8WD^3n/(Gd^4)$, so $n = Gd^4\delta/(8WD^3)$.

$n = 80\,000 \times 10^4 \times 25/(8 \times 800 \times 60^3) = 2 \times 10^{10}/1.3824 \times 10^9 = 14.47$

Turns must be whole numbers, so provide **15 active turns**.

Exam tip: Rounding the turns up makes the spring slightly softer (more deflection), which is usually the safe side for a deflection requirement; stress is unaffected by n .

Q14 Answer: (A) 1.253

Concept: $K_W = (4C - 1)/(4C - 4) + 0.615/C$ combines the curvature and direct-shear corrections.

$K_W = 23/20 + 0.615/6 = 1.150 + 0.1025 = \mathbf{1.253}$

Exam tip: 1.083 is the direct-shear factor alone ($1 + 0.5/C$) and 1.150 is only the first term. A smaller index C means a larger correction.

Q15 Answer: (B) 9k

Concept: $k = Gd^4/(8D^3n) \propto 1/n$. Each piece has $n/3$ turns, so each piece has stiffness $3k$.

Three pieces in parallel: $k_{eq} = 3k + 3k + 3k = \mathbf{9k}$

Exam tip: The same rule holds for any spring or elastic bar: stiffness is inversely proportional to the active length.

What next?

Day 11 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



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